Dhirubhai Ambani University

2024 01419 <202401419@dau.ac.in>

Fwd:

2024 01165 <202401165@dau.ac.in> Fri, Sep 25, 2026 at 11:07 PM
To: 202401419@dau.ac.in

---------- Forwarded message ---------
From: **2024 01165** <202401165@dau.ac.in>
Date: Fri, Sep 25, 2026 at 11:04 PM
Subject:
To: <202401077@dau.ac.in>

Phase 1 — Evaluation Metrics

Category	Metric	What it measures	Priority
Correctness	File Transfer Success Rate	% of client requests for which all expected files are correctly received	**High**
	File Integrity	Whether received files match the originals	**High**
	Topic Handling	Correct topic → directory mapping and response	**High**
Performance	**Total Completion Time**	Time taken for all clients to complete their requests	**Very High**
	Aggregate Throughput	Total bytes transferred / total elapsed time	**Very High**
	Per-client Throughput	Bytes received by a client / its transfer time	High
	Scalability	How throughput changes as number of clients increases	**Very High**
Concurrency	Speedup	T_{single}/T_{multi} for the same workload	**Very High**
	Efficiency	Speedup / number of clients	Medium
Resource/behavior	CPU utilization	Server CPU consumption under different loads	Medium
	Memory usage	Server memory consumption	Low/Medium

Memory usage Server memory consumption Low/Medium

The 4 metrics I would make the core evaluation

If this were my course, I would make students report these four prominently:

1. Total Completion Time
For N clients:

$$T_{completion} = t_{last\ client\ completion} - t_{experiment\ start}$$

Example:

Clients	Single-threaded	Multi-threaded
1	2.1 s	2.1 s
2	4.2 s	2.3 s
4	8.4 s	4.5 s
8	16.7 s	8.7 s

This directly demonstrates the benefit of concurrency.

---

2. Aggregate Throughput
This is **the most important metric in your assignment**, because the PDF explicitly asks for aggregate file transfer rate. IT305-lab-socket-programming-pr...

$$\boxed{Throughput = \frac{\text{Total bytes transferred}}{\text{Total completion time}}}$$

Usually report it in:

$$Throughput = \frac{500\,MB}{10\,s} = 50\,MB/s$$

Plot:

X-axis: Number of clients
Y-axis: Aggregate throughput (MB/s)

with two curves:

- Single-threaded
- Multi-threaded

That graph should be one of your main Phase 1 results.

---

3. Speedup
This makes your comparison much more meaningful:

$$\boxed{Speedup(N) = \frac{T_{single}(N)}{T_{multi}(N)}}$$

For example:

$$T_{single} = 12s, \qquad T_{multi} = 7s$$

$$Speedup = \frac{12}{7} = 1.71\times$$

You can plot:

Number of clients vs Speedup

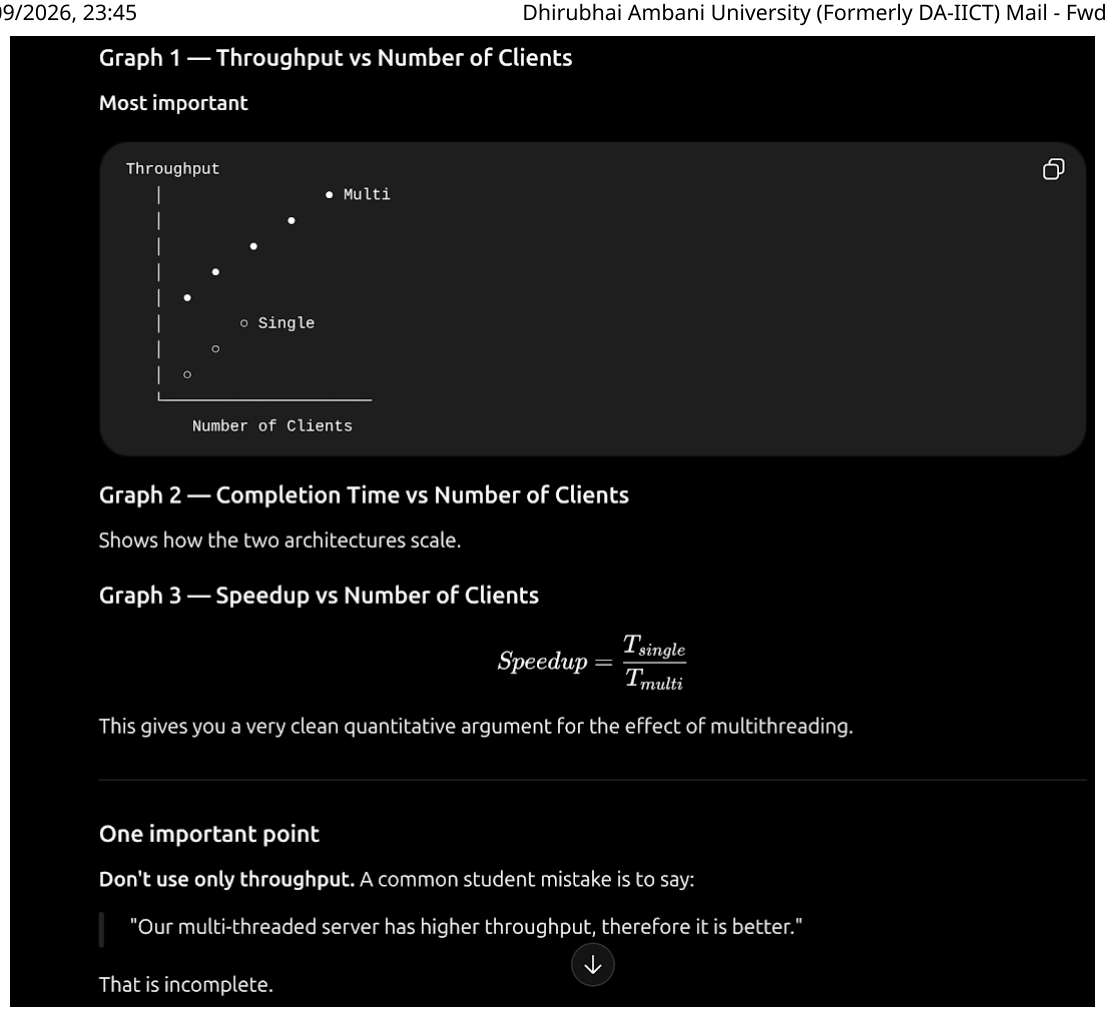
This tells the evaluator how much benefit you actually obtained from multithreading.

Share

Recommended final results table

# Clients	Server	Total Time (s)	Total Bytes (MB)	Aggregate Throughput (MB/s)	Avg Client Throughput (MB/s)	Success Rate
1	Single					
1	Multi					
2	Single					
2	Multi					
4	Single					
4	Multi					
8	Single					
8	Multi					

Then generate **three graphs**:



That is incomplete.

You should analyze **why throughput changes as clients increase** and identify where the multi-threaded implementation stops scaling. The assignment specifically asks you to analyze and comment on the performance of the two implementations.

So for Phase 1, my professor-level evaluation framework would be:

And I would keep **fault rate, retransmitted bytes, redundant bytes, checkpoint overhead, etc. out of Phase 1** because those are explicitly part of the Part II evaluation.